

Assignment 7

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Preliminaries

Package Loading

```
library(httr)
```

```
## Warning: package 'httr' was built under R version 3.2.5
```

```
library(dplyr)
```

```
## Warning: package 'dplyr' was built under R version 3.2.5
```

```
##
```

```
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      filter, lag
```

```
## The following objects are masked from 'package:base':
```

```
##
```

```
##      intersect, setdiff, setequal, union
```

```
library(purrr)
```

```
## Warning: package 'purrr' was built under R version 3.2.5
```

```
##
```

```
## Attaching package: 'purrr'
```

```
## The following objects are masked from 'package:dplyr':
```

```
##
```

```
##      contains, order_by
```

```
library(stringr)
```

```
library(ggplot2)
```

```
## Warning: package 'ggplot2' was built under R version 3.2.5
```

```
library(data.table)
```

```
## Warning: package 'data.table' was built under R version 3.2.5
```

```
## -----
```

```
## data.table + dplyr code now lives in dtplyr.
```

```
## Please library(dtplyr)!
```

```
## -----
```

```
##
```

```
## Attaching package: 'data.table'
```

```
## The following object is masked from 'package:purrr':
```

```
##
```

```
##      transpose
```

```
## The following objects are masked from 'package:dplyr':
```

```
##
```

```
##      between, first, last
```

Assignment Part I

1) *Answer to Problem 1*

Sample space: $\Omega = \{\text{win-win, win-lose, win-tie, tie-win, tie-lose, tie-tie, lose-win, lose-lose, lose-tie}\}$ Let X be the random variable of the number of points the team earns. Then we have

$X = 4$ if $w = \text{win-win}$

$X = 3$ if $w = \text{win-tie, tie-win}$

$X = 2$ if $w = \text{win-lose, lose-win, tie-tie}$

$X = 1$ if $w = \text{tie-lose, lose-tie}$

$x = 0$ if $w = \text{lose-lose}$

$$P(X = 4) = 0.4 \times 0.5 \times 0.7 \times 0.5 = 0.07$$

$$P(X = 3) = 2 \times (0.4 \times 0.5 \times 0.7 \times 0.5) = 0.14$$

$$P(X = 2) = 0.4 \times 0.5 \times 0.3 + 0.6 \times 0.7 \times 0.5 + 0.4 \times 0.5 \times 0.7 \times 0.5 = 0.34$$

$$P(X = 1) = 0.4 \times 0.5 \times 0.3 + 0.6 \times 0.5 \times 0.7 = 0.27$$

$$P(X = 0) = 0.6 \times 0.3 = 0.18$$

As a result, the PMF $p_X(x)$ is

$$p_X(x) = \begin{cases} 0.07 & \text{if } x = 4 \\ 0.14 & \text{if } x = 3 \\ 0.34 & \text{if } x = 2 \\ 0.27 & \text{if } x = 1 \\ 0.18 & \text{if } x = 0 \\ 0 & \text{otherwise.} \end{cases}$$

2) *Answer to Problem 2*

- (a) Fischer can win the match by winning the first game, or win the second game after a draw, or win the third game after two successive draws, or... win the tenth game after nine successive draws. Therefore, the probability P that Fischer wins the match is. The probability that Fischer wins the game is 0.5714252.

$$\sum_{i=1}^9 (0.3)^i \times 0.4 = 0.5714252$$

```
x<-0:9
sum((0.3^x)*0.4)
```

```
## [1] 0.5714252
```

- (b) The game can be ended by three situations: first, someone wins before the 10th game; second, all the 10 games are draws; third, the 1st to 9th game are draws when, someone wins the 10th game. In the first case, the duration of the game (d) can be d (someone wins the dth game); the second and third case, the duration of game is 10. Equivalently, if all the first 9 games are draws, no matter what happen in the 10th game, the duration is 10.

$$P(D = d) = \begin{cases} (0.3)^d \cdot 0.7 & d = 0, 1, 2, \dots, 8 \\ (0.3)^9 & d = 10 \\ 0 & \text{otherwise} \end{cases}$$

3) *Answer to Problem 3*

- a) a=28 and E[X]=0.

We know that the possibility that all events happen is 1, thus:

$$\begin{aligned} \sum_{x=-3}^3 \frac{x^2}{a} &= 1 \\ \frac{1}{a} \sum_{-3}^3 x^2 &= 1 \\ \frac{1}{a} \times 28 &= 1 \\ a &= 28 \end{aligned}$$

Thus, the E[X] is:

$$\begin{aligned} E[X] &= \sum_{-3}^3 \frac{x^2}{28} \cdot x \\ &= 0 \end{aligned}$$

```
x <- -3 : 3
sum(x^2)
```

```
## [1] 28
```

```
sum((x^3)/28)
```

```
## [1] 0
```

b) What is the PMF of the random variable Z

We know that $E[X]=0$, thus $Z = X^2$. Since $x=-3,-2,-1,0,1,2,3$, Z could be $0,1,4,9$.

if $Z \in \{1, 4, 9\}$, we have

$$p_Z(z) = p_X(\sqrt{z}) + p_X(-\sqrt{z}) = \frac{z}{28} + \frac{z}{28} = \frac{z}{14}$$

if $Z = 0$

$$p_Z(z) = p_X(x = 0) = \frac{0^2}{28} = 0$$

$p_Z(z) = 0$ otherwise

Therefore, the PMF of the random variable Z is

$$p_Z(z) = \begin{cases} \frac{z}{14} & \text{for } z \in \{1, 4, 9\} \\ 0 & \text{otherwise.} \end{cases}$$

c) Find the variance of X . The variance of X is 7.

$$\begin{aligned} \text{var}(X) &= E[X^2] - (E[X])^2 \\ &= E[Z] - 0 \\ &= \sum_{z=1,4,9} \frac{z}{14} \cdot z \\ &= 7 \end{aligned}$$

d) find the variance of X . The variance of X is 7.

$$\begin{aligned} \text{var}(X) &= \sum_x (x - E[X])^2 p_X(x) \\ &= \sum_x x^2 \frac{x^2}{28} \\ &= \sum_{x=-3}^3 \frac{x^4}{28} \\ &= 7. \end{aligned}$$

```
x <- -3 : 3
sum((x^4)/28)
```

```
## [1] 7
```

4) *Answer to Problem 4*

The situation in the question is a geometric probability distribution, which is the probability distribution of the number X of Bernoulli trials needed to get one success, supported on the set $\{1, 2, 3, \dots\}$. If the probability of success on each trial is p , then the probability that the k th trial is the first success is :

$$\Pr(X = k) = (1 - p)^{k-1} p$$

The mean of the geometric distribution is $\frac{1}{p}$, and the variance of it is $\frac{1-p}{p^2}$

$$\begin{aligned} \sum_{k=1}^{\infty} f_X(k) &= \sum_{k=1}^{\infty} (1 - p)^{k-1} p \\ &= p \sum_{j=0}^{\infty} (1 - p)^j \\ &= p \frac{1}{1 - (1 - p)} \\ &= 1 \end{aligned}$$

We can now compute $E[X]$:

$$\begin{aligned}
E[X] &= \sum_{k=1}^{\infty} k \times (1-p)^{k-1} p \\
&= p \left(\sum_{k=1}^{\infty} (1-p)^{k-1} + \sum_{k=2}^{\infty} k(1-p)^{k-1} + \sum_{k=3}^{\infty} (1-p)^{k-1} + \dots \right) \\
&= p \left(\left(\frac{1}{p} \right) + \frac{1-p}{p} + \frac{(1-p)^2}{p} + \dots \right) \\
&= 1 + (1-p) + (1-p)^2 + \dots \\
&= \frac{1}{p}
\end{aligned}$$

5) **Answer to Problem 5**

$$\begin{aligned}
p_Y(y=1) &= P(X \leq \frac{1}{3}) = \frac{1}{3} \\
p_Y(y=2) &= P(\frac{1}{3} \leq X \leq 1) = \frac{2}{3}
\end{aligned}$$

The expected value of Y is:

$$\begin{aligned}
\sum_{y=1,2} y p_y &= 1 \times \frac{1}{3} + 2 \times \frac{2}{3} \\
&= \frac{5}{3}
\end{aligned}$$

We know that $X \sim \text{Uniform}(0,1)$. According to the expected value rule:

$$\begin{aligned}
E[Y] &= E[g(x)] = \int_{-\infty}^{\infty} g(x) f_X(x) dx \\
&= \int_0^{\frac{1}{3}} g(x) f(x) dx + \int_{\frac{1}{3}}^1 g(x) f(x) dx \\
&= \int_0^{\frac{1}{3}} dx + \int_{\frac{1}{3}}^1 2 dx \\
&= \frac{5}{3}
\end{aligned}$$

6) **Answer to Problem 6**

a) The normalization condition is that:

$$P(-\infty < x < \infty) = \int_{-\infty}^{\infty} P(x) dx \equiv 1$$

In this case:

$$\begin{aligned}
 & \int_{-\infty}^{\infty} \frac{\lambda}{2} e^{-\lambda|x|} dx \\
 &= \frac{\lambda}{2} \int_{-\infty}^{\infty} e^{-\lambda|x|} dx \\
 &= \frac{\lambda}{2} \left(\int_{-\infty}^0 e^{\lambda x} dx + \int_0^{\infty} e^{-\lambda x} dx \right) \\
 &= \frac{\lambda}{2} ((1-0) - (0-1)) \\
 &= 1
 \end{aligned}$$

So f_X satisfies the normalization condition.

b) mean is 0

$E(X)$ is 0 because the PDF is symmetric around 0.

c) variance is $\frac{\lambda^2}{2}$

$$\begin{aligned}
 E(X^2) &= \int_{-\infty}^{\infty} x^2 \frac{\lambda}{2} e^{-\lambda|x|} dx \\
 &= 2 \times \frac{1}{2} \times \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx = \frac{2}{\lambda^2} \\
 \text{var}(X) &= E(X^2) - \{E(x)\}^2 \\
 &= \frac{2}{\lambda^2} - 0 \\
 &= \frac{2}{\lambda^2}
 \end{aligned}$$

7) *Answer to Problem 7*

(a) PDF of X

According to the Law of Total Probability,

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + \dots + P(A|E_n)P(E_n)$$

The PDF of X is:

$$\begin{aligned}
 f_X(x) &= P(X \leq x) \\
 &= pP(Y \leq x) + (1-p)P(Z \leq x) \\
 &= pf_Y x + (1-p)f_Z(x)
 \end{aligned}$$

b) CDF of the two-sided exponential random variable

If $x < 0$, the CDF is

$$\begin{aligned} F_X(x) &= \int_{-\infty}^x p\lambda e^{x\lambda} \\ &= pe^{\lambda x} \end{aligned}$$

If $x \geq 0$, the CDF is:

$$\begin{aligned} F_X(x) &= \int_{-\infty}^0 p\lambda e^{x\lambda} + \int_0^x (1-p)\lambda e^{-\lambda x} \\ &= pe^0 + (1-p)(-e^{-\lambda x} \big|_0^x) \\ &= p + (1-p)(1 - e^{-\lambda x}) \\ &= 1 - (1-p)e^{-\lambda x} \end{aligned}$$

So the CDF of the whole function is: $F_X(x) = p \cdot F_Y(x) + (1-p) F_Z(x) = \begin{cases} p \cdot e^{\lambda x} & \text{if } x < 0 \\ 1 - (1-p)e^{-\lambda x} & \text{if } x \geq 0 \end{cases}$

8) *Answer to Problem 8*

$X \sim \text{Normal}(0,1)$, $Y \sim \text{Normal}(1,4)$

a) X is a standard normal distribution, we can refer to the normal table directly.

According to the table, we have $P(X \leq 1.5) = 0.9332$. Also, the graph is symmetric about 0, so we know that $P(X \leq -1) = P(X \geq 1) = 1 - P(X \leq 1) = 1 - 0.8413 = 0.1587$

b)

$Z = \frac{Y-1}{2} = \frac{Y-\mu}{\sigma}$, so it is a standard normal distribution, the proof is shown below, and the PDF of it is $\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y)^2}$

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P\left(\frac{Y-\mu}{\sigma} \leq z\right) \\ &= P(Y \leq \mu + \sigma z) \\ &= \int_{-\infty}^{\mu + \sigma z} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2} dy \\ &= \int_{-\infty}^{\mu + \sigma z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y)^2} dy \end{aligned}$$

Differentiate the CDF we can get the PDF.

c) We transfer the distribution into standard normal distribution, then consult to the normal distribution table

$$\begin{aligned}
F_Y(y) &= P(-1 \leq Y \leq 1) \\
&= P(-1 \leq \frac{Y-1}{2} \leq 0) \\
&= P(-1 \leq x \leq 0) \\
&= P(x \leq 1) - P(x \leq 0) \\
&= 0.8413 - 0.5 \\
&= 0.3413
\end{aligned}$$

X in a standard normal random variable in the functions above

- 9) Display the prevalence of mayonnaise, sauerkraut, and dill in the American Pickle Packer Association recipes.

In this case, I interpret the prevalence of mayonnaise, sauerkraut, and dill in recipes not only as using exactly "mayonnaise", "saurkraut", and "dill", but also their variants such as "low-fat mayonnaise". I think no matter what type of mayonnaise it is, it is mayonnaise.

```
# Organized Data
PickleRecipesSite <- GET("http://ilovepickles.org/book/export/html/2")
PickleRecipesText <- content(PickleRecipesSite, as = "text")

PickleRecipesText <- str_replace_all(PickleRecipesText, "[\n]", " ")

ingredients <- unlist(str_extract_all(PickleRecipesText, "What You Need:&nbsp;</div>.+?</div>"))
ingredients <- str_replace_all(ingredients, "<.+?>", ", ")

ingredientslist <- str_split(ingredients, ",")
trimingredients <- map(ingredientslist, str_trim)
rep(c("a", "b"), c(1, 3))

## [1] "a" "b" "b" "b"

recipenumber <- rep(1:length(trimingredients), map(ingredientslist, length))
dfpickles <- data.frame(recipenumber = recipenumber, ingredient = unlist(trimingredients))

dfpickles <- dfpickles %>% filter(!ingredient %in% c("What You Need:&nbsp;", "&nbsp;",
"")) %>% mutate(ingredient = tolower(ingredient),

#Used str_detect to detect
dfpicklesig <- mutate(dfpickles,
  my=str_detect(ingredient, "mayonnaise"),
  sk=str_detect(ingredient, "sauerkraut"),
  di=str_detect(ingredient, "dill"),
  egg=str_detect(ingredient, "egg"))

#Used group_by to count
ingredientcount<-dfpicklesig%>%group_by(egg)%>%mutate(mayonnaise=length(my[my==TRUE]))%>%
  mutate(sauerkraut=length(sk[sk==TRUE]))%>%
```

```

mutate(dill=length(di[di==TRUE]))

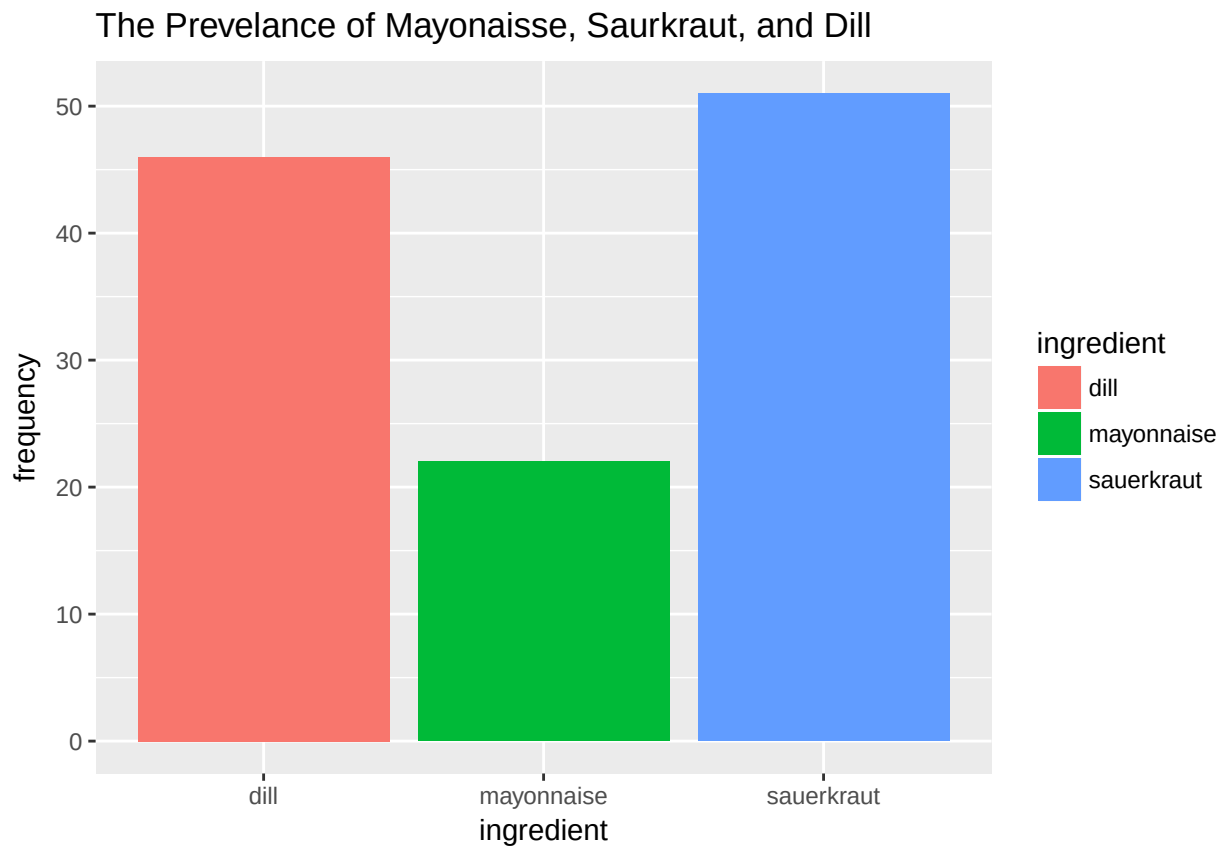
# subset data and transpose the data frame
ingredientcount2<-ingredientcount[2,c(7,8,9)] # keep the rows and columns I need
ingredientcount3<-as.data.frame(t(ingredientcount2)) # transpose the data.frame
setDT(ingredientcount3, keep.rownames = TRUE)[] #transfer rownames to the first column

##           rn V1
## 1: mayonnaise 22
## 2: sauerkraut 51
## 3:          dill 46

names(ingredientcount3)<-c("rn"="ingredient", "V1"="frequency") # rename the variables

# use ggplot to plot
ggplot(ingredientcount3, aes(x = ingredient , y = frequency,fill=ingredient)) + geom_bar(stat = "identity")
labs(title="The Prevalance of Mayonaisse, Saurkraut, and Dill")

```



10) Use R to evaluate Isaac Newton's claim (use simulation).

The result of the simulation shows that the probability of experA is the highest (0.6633). According to this simulation (which gives us an approximation of a probability), the Newton's claim is right.

```
die<-c(1:6)
exB<-function(d,s,n){
  toss <-sample(d,s,replace=TRUE)
  atleast<-ifelse(length(which(toss==6))>=n,1,0)
  return(atleast)
}
```

```
experA<-replicate(10000,exB(die,6,1))
table(experA)/length(experA)
```

```
## experA
##      0      1
## 0.3367 0.6633
```

```
experB<-replicate(10000,exB(die,12,2))
table(experB)/length(experB)
```

```
## experB
##      0      1
## 0.378 0.622
```

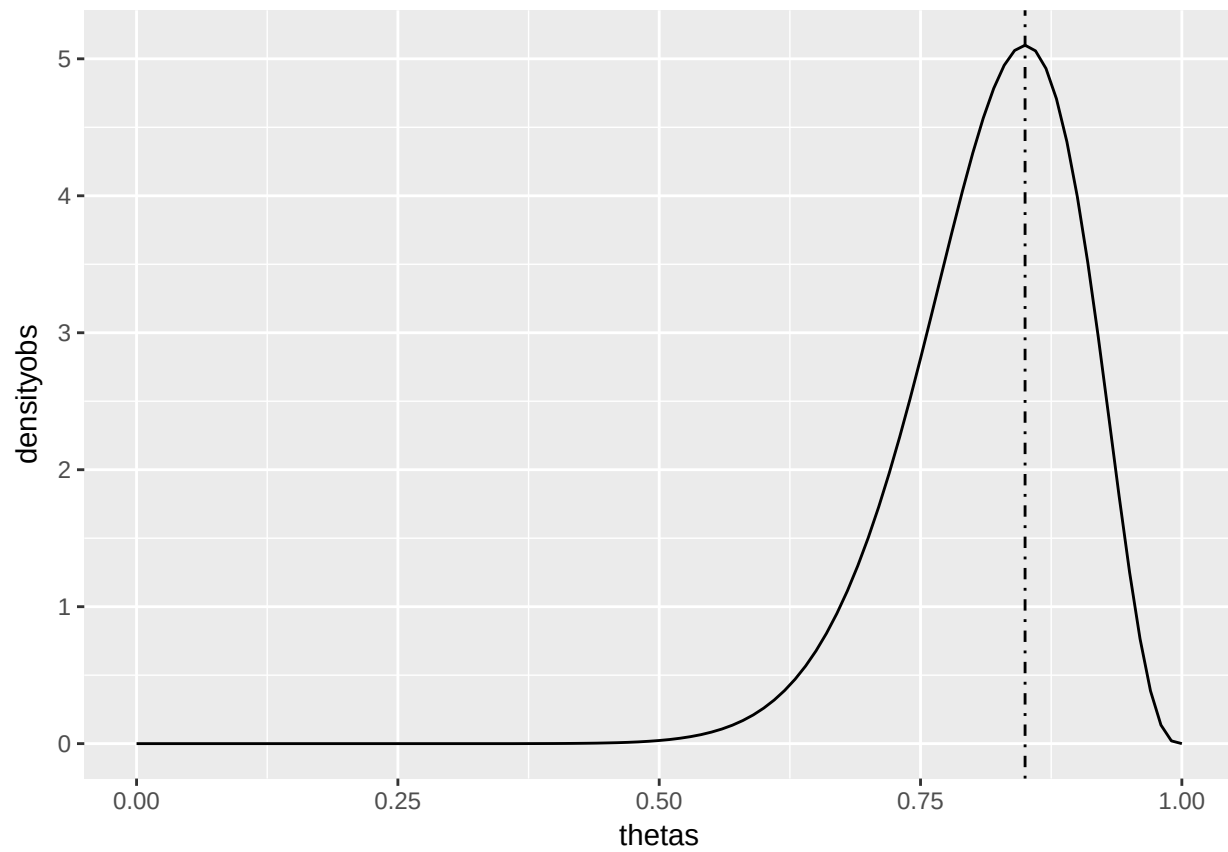
```
experC<-replicate(10000,exB(die,18,3))
table(experC)/length(experC)
```

```
## experC
##      0      1
## 0.4034 0.5966
```

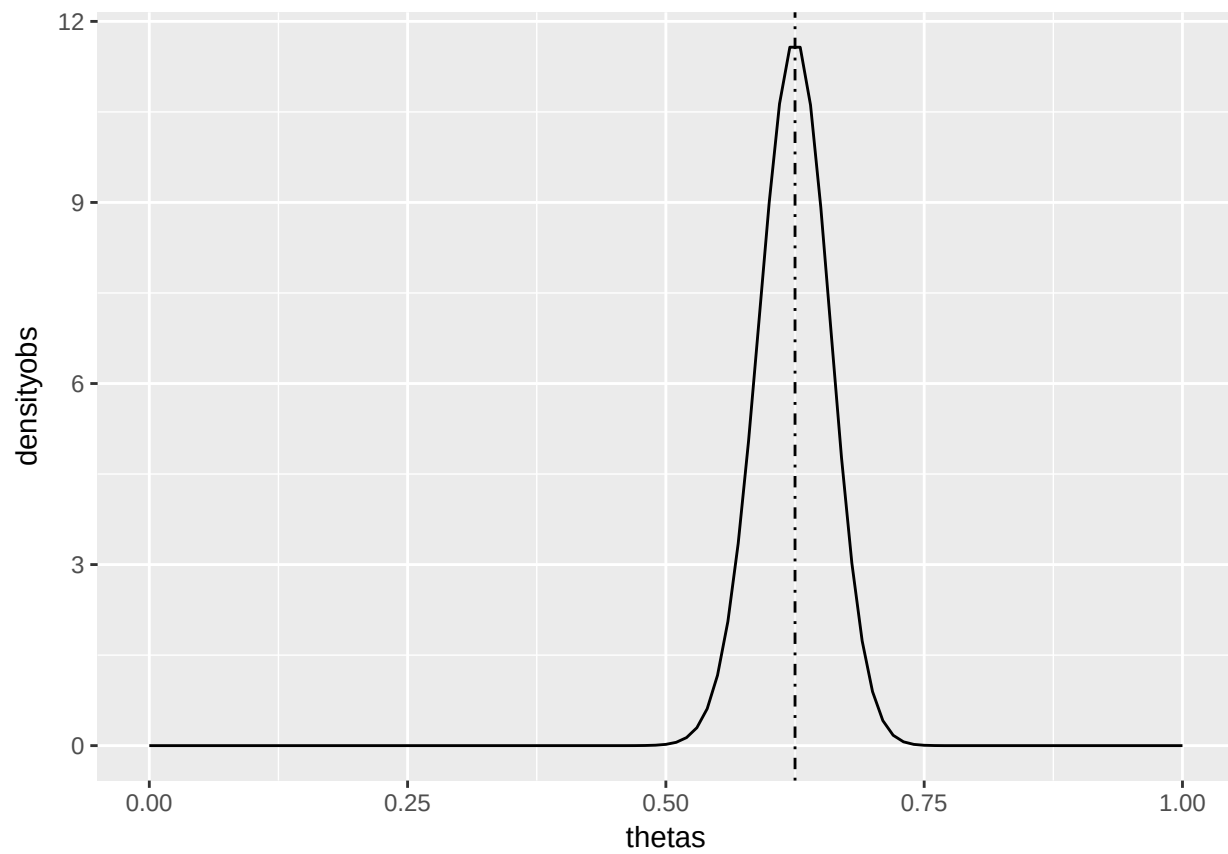
11)

- a) Use the Beta distribution to plot the posterior probability density function of an individual supporting Trump if we observe his support at 17 of 20, 125 of 200, or 1082 of 2000.

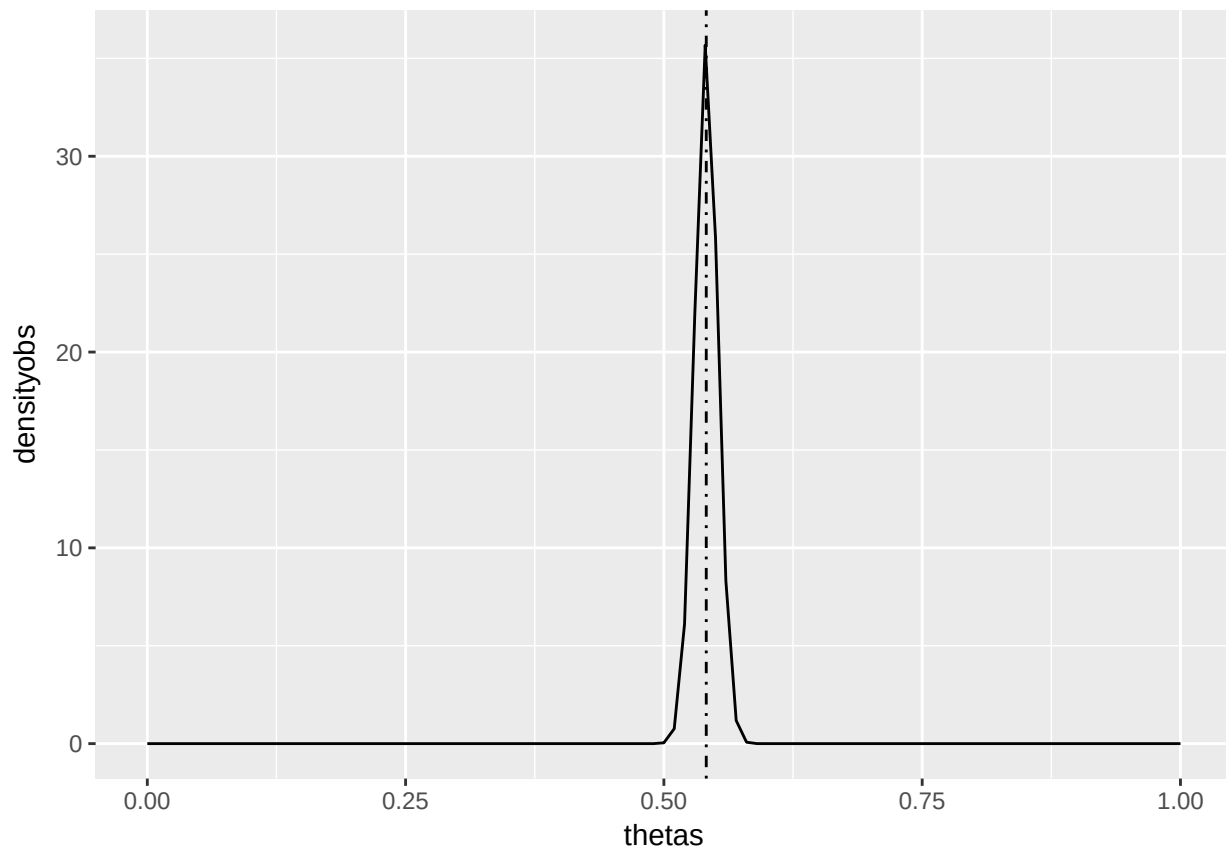
```
simposterior1<-data.frame(thetas= seq(0,1,by=.01),
                          densityobs=dbeta( seq(0,1,by=.01),17+1,20-17+1))
ggplot(simposterior1,aes(thetas, densityobs))+geom_freqpoly(stat="identity")+
  geom_vline(xintercept =17/20,linetype = "dotted")
```



```
simposterior2<-data.frame(thetas= seq(0,1,by=.01),  
                           densityobs=dbeta( seq(0,1,by=.01),125+1,200-125+1))  
ggplot(simposterior2,aes(thetas, densityobs))+geom_freqpoly(stat="identity")+  
  geom_vline(xintercept =125/200,linetype = "dotdash")
```



```
simposterior3<-data.frame(thetas= seq(0,1,by=.01),  
                           densityobs=dbeta( seq(0,1,by=.01),1082+1,2000-1082+1))  
ggplot(simposterior3,aes(thetas, densityobs))+geom_freqpoly(stat="identity")+  
  geom_vline(xintercept =1082/2000,linetype = "dotdash")
```



b) Which of these should Kellyanne report to the President? Which is most heartening for the President? Which is 'fake news'?

First, I think Kellyanne should report the third one, since it contains the most certain information. From the shape of the three graphs, the first is the "fattest" one and the third is the "thinnest" one. The "fatter" the graph, the more uncertain it is.

The first sample 17 of 20 is the most heartening for the President because its posterior θ is the highest. It means that Trump has more supporters.

From the perspective of Trump, the last sample of 1082 of 2000 is 'fake news' because the result is the least welcoming to him. Statistically, however, the first sample 17 of 20 is fake news because the sample is too small to reflect the fact in the population. According to the Central Limit Theorem, given a sufficiently large sample size from a population with a finite level of variance, the mean of all samples from the same population will be approximately equal to the mean of the population. The sample in graph 1 is too small to be approximately equal to the mean of the population.